

Question 1

Solve x . Answer in interval notation where applicable.

1. $|3 - 2x| \leq 7$
2. $|3x - 4| > 8$
3. $\frac{x+1}{2+x} \leq 0$
4. $\frac{x+2}{|x-1|} > 0$
5. $\ln(3x+1) - \ln x = -2$

Question 2

Find the domain of f

1. $f(x) = \frac{\sqrt{1-x}}{x}$.
2. $f(x) = |\tan x|$, $x \in (0, 2\pi)$
3. $f(x) = \ln(x^2 - 1)$.

Question 3

1. $\sin(10\pi) =$
2. $\tan(\frac{4\pi}{3}) =$
3. $\sec(\frac{5\pi}{2}) =$
4. $\cos x = -\frac{\sqrt{3}}{2} \Rightarrow x =$
5. $\cos x > -\frac{\sqrt{3}}{2}, x \in [0, 2\pi] \Rightarrow x \in$

Question 4

Determine $f^{-1}(x)$ and the domain f^{-1}

1. $f(x) = x^3 + 2$
2. $f(x) = 2e^{-x}$

Question 5

Compute $(f \circ g)(x)$, $(g \circ f)(x)$, $(\frac{f}{g})(x)$ and $(fg)(x)$ determine the domains of each.

$$f(x) = \frac{x}{3+2x}, g(x) = 8+5x$$

Question 6

Sketch the graph of

1. $y = 10 - e^{-3x}$
2. $y = -\ln(x-2)$
3. $y = \sqrt{2x-x^2}$

Worksheet: Week 4

Question 1

1. $|3-2x| \leq 7$

$\Rightarrow x \in [-2, 5]$

2. $|3x-4| > 8$

$\Rightarrow x \in (-\infty, -\frac{4}{3}) \cup (4, \infty)$

3. $\frac{x+1}{2+x} \leq 0$

$\Rightarrow x \in (-2, -1]$

4. $\frac{x+2}{|x-1|} > 0$

$\Rightarrow x \in (-2, 1) \cup (1, \infty)$

5. $\ln(3x) - \ln(x) = -2$

$\Rightarrow x = \frac{e^2}{1-3e^2}$

Question 2

1. $f(x) = \frac{\sqrt{1-x}}{x}$

$D_f = (-\infty, 0) \cup (0, 1]$

2. $f(x) = |\tan x|, x \in (0, 2\pi)$

$D_f = (0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \frac{3\pi}{2}) \cup (\frac{3\pi}{2}, 2\pi)$

3. $f(x) = \ln(x^2-1)$

$D_f = (-\infty, -1) \cup (1, \infty)$

Question 3

1. 0 2. $\sqrt{3}$ 3. Not def.

4. $x = \frac{5\pi}{6} + 2k\pi$ or

$x = \frac{7\pi}{6} + 2k\pi, k \in \mathbb{Z}$

5. $x \in [0, \frac{5\pi}{6}) \cup (\frac{7\pi}{6}, 2\pi]$

Question 4

1. $f(x) = x^3 + 2$

$f'(x) = (x-2)^{1/3}$

$D_f = R_{f^{-1}} = \mathbb{R}$

$R_{f^{-1}} = D_{f^{-1}} = \mathbb{R}$

2. $f(x) = 2e^{-x}$

$f'(x) = -\ln(\frac{x}{2})$

$D_f = R_{f^{-1}} = \mathbb{R}$

$R_f = D_{f^{-1}} = (0, \infty)$

Question 5

$f(x) = \frac{x}{3+2x}, g(x) = 8+5x$

$(f \circ g)(x) = \frac{8+5x}{19+10x}, x \neq -\frac{19}{10}$

$D = \mathbb{R} \setminus \{-\frac{19}{10}\}$

$(g \circ f)(x) = 8 + \frac{5x}{3+2x}, x \neq -\frac{3}{2}$

$D = \mathbb{R} \setminus \{-\frac{3}{2}\}$

$(\frac{f}{g})(x) = \frac{x}{(3+2x)(8+5x)}$

$D = \mathbb{R} \setminus \{-\frac{3}{2}, -\frac{8}{5}\}$

$(fg)(x) = \frac{x(8+5x)}{3+2x}$

$D = \mathbb{R} \setminus \{-\frac{3}{2}\}$

Question 6:

